Preimplantation genetic testing to evaluate for mitochondrial deoxyribonucleic acid disease and aneuploidy: a two-birds-with-one-stone approach

Mitochondrial deoxyribonucleic acid (mtDNA) mutations, although rare, can result in significant disruptions in the normal process of oxidative phosphorylation within cells (1). In humans, the phenotypes of maternally inherited mtDNA mutations are extremely variable and can clinically manifest from infancy to adulthood in a single organ or as part of a multisystem disease process (2). It has been well recognized that mitochondrial cytopathies can result in severe, often lethal, degenerative cardiovascular problems, neurologic disorders, or musculoskeletal abnormalities. However, treatment options for individuals affected by mtDNA mutations are generally inadequate and primarily palliative.

Because of the significant limitations of the available treatment options, a parental desire to avoid having children affected by mtDNA mutations is understandable. Unfortunately, individuals who are at a risk of transmitting mtDNA mutations to their offspring are often left with limited reproductive options. Pursuing in vitro fertilization using donor oocytes prevents the maternal transmission of mtDNA mutations, but this option precludes the possibility of a genetic link between the mother and child. The options of nuclear genome transfer from a pronuclear-stage zygote of an affected woman into an enucleated donor zygote or the transfer of the spindle from an unfertilized metaphase-II oocyte of an affected woman into an enucleated donor oocyte have also been described (3). Although interesting, the clinical application of the latter options has been limited because of technical, legal, and ethical challenges associated with the "three-parent in vitro fertilization" approach.

Spath et al. (4) described another option: a strategy involving embryo biopsy and preimplantation genetic testing (PGT) to simultaneously screen for mtDNA disease and aneuploidy. This option allows individuals at a risk of transmitting mtDNA disease to maintain a biological link with the offspring while theoretically ensuring the transfer of a euploid embryo free of mtDNA mutations. In this case series of 4 patients and subsequent laboratory validation of the testing platform, Spath et al. (4) provided a reassuring report of the accuracy of the mtDNA mutation quantification test. Additionally, the investigators highlighted the fact that strategies combining PGT for mtDNA disease with PGT for aneuploidy are likely to be superior to approaches that only consider testing for mtDNA disease because in the study by Spath et al. (4), approximately half of the embryos tested were aneuploid. Although the rates of aneuploidy change based on patient age, the investigators' point was salient. The investigators appropriately argued that testing for aneuploidy allows clinicians to avoid the physical and emotional consequences of aneuploid embryo transfers, such as miscarriages, negative pregnancy tests, and deliveries of children with chromosomal aneuploidies.

One of the greatest strengths of the study by Spath et al. (4) is the rigorous approach the investigators took for validating the PGT method. Prior studies have noted that mtDNA mutations implicated in mtDNA disease often show heteroplasmy which denotes the coexistence of wild-type and pathogenic variants of mtDNA in a single cell (5). This concept of heteroplasmy complicates the inheritance of mtDNA mutations. Essentially, because of a genetic bottleneck during oogenesis, the proportion of pathogenic variant mtDNA inherited from one generation to the next displays wide variability, and clinical symptoms are present only when a critical mutant load threshold is exceeded (5).

In the study by Spath et al. (4), validation was performed using genomic DNA from buccal swabs and/or blood samples of probands, affected children, and controls. This validation process tested the accuracy and sensitivity of sample mixtures derived from homoplasmic wild-type and homoplasmic variant sample mixtures. Different levels of mutant mtDNA were observed and compared with levels previously established by reference laboratories. Although no muscle DNA samples were analyzed, it is encouraging to note that the PGT method described was able to accurately assess these varying mutant loads. This ability to distinguish exact mutant mtDNA loads is what allows for the correct identification of an embryo based on a load either above or below the critical load threshold of mutant mtDNA required to express the disease.

The work by Spath et al. (4) was the first to describe the successful application of a next-generation sequencing–based PGT approach for both mtDNA disease and aneuploidy. The protocol used by Spath et al. (4) demonstrated strength in its ability to accurately test samples consisting of only a few cells—a necessity for a test applied to embryo biopsy samples. Because the protocol involved whole-genome amplification (WGA), it was also important for the investigators to ensure that the performance of WGA before the amplification of mutant loci did not distort the variant ratios of the samples. This essential step was performed, and one of the highlights of the study by Spath et al. (4) was the fact that the mutant loads obtained after direct mutation site amplification and after WGA before mutation site amplification were nearly identical to the expected ratios.

This study serves as an excellent addition to the growing body of literature related to mtDNA mutations. One of the main limitations, however, is the fact that only 4 patients were evaluated, and the specific mtDNA mutations present in these patients were heterogeneous. The study's small sample size resulted in only one patient having a delivery after an embryo transfer. This emphasizes the fact that further exploration of the clinical applications of this technology is necessary. Additionally, because this study analyzed biopsy samples obtained at the cleavage and blastocyst stages, a future study should exclusively focus on an analysis of blastocyst-stage biopsies.

The work by Spath et al. (4) offers a new strategy to patients, who might significantly benefit from additional

fertility options. As the investigators rightfully noted, this study provides useful information for patient counseling. Furthermore, many prior studies evaluating PGT for mtDNA mutations did not account for the effect of aneuploidy on subsequent clinical outcomes; therefore, this study is unique in its intent to generate PGT results that provide as much relevant information for patients as possible. Although the true success of this protocol will be measured based on healthy live births in the future, Spath et al. (4) provided a novel approach that can serve as a guide for appropriate reproductive planning in patients affected by mtDNA mutations.

Brent M. Hanson, M.D.[a,b]
Emre Seli, M.D.[a,c]

[a] IVI-Reproductive Medicine Associates of New Jersey, Basking Ridge, New Jersey, [b] Sidney Kimmel Medical College at Thomas Jefferson University, Philadelphia, Pennsylvania, and [c] Yale University School of Medicine, New Haven, Connecticut

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